Spencer Beer

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Summary

Experienced in automotive engineering, including development of automotive ADAS, operating systems, cybersecurity penetration-testing tools, and reverse engineering embedded automotive firmware. Specialized in machine learning, embedded systems, IoT, and systems security.

Education

Colorado State University | GPA 3.6 — Fort Collins, CO
Master of Science in Systems Engineering, Embedded and Heavy Vehicle Cyber Security — 08/2024 - 05/2026

Colorado State University | GPA 3.6 — Fort Collins, CO
Bachelor of Science in Computer Engineering, Embedded Systems, IoT, and Machine Learning — 08/2020 - 05/2024

Technical Skills

Languages & HDLs: C, C++, Python, Java, JavaScript, MATLAB, SystemC; Assembly (ARM/MIPS/x86/XTensa); Verilog, VHDL; SQL; HTML/CSS (Bootstrap); Bash & PowerShell
AI / Data Frameworks: TensorFlow, PyTorch, scikit-learn, OpenCV, MCP, llama-cpp; Parallel programming (CUDA, OpenCL, OpenMP)
Embedded & Hardware: Yocto, Buildroot, FreeRTOS; STM32, ESP32, Raspberry Pi, BeagleBone; Communication protocols (SPI, I²C, CAN, LIN, Ethernet, UART, RS-485/232, JTAG/SWD, MQTT); Soldering (incl. SMT), PCB debug (Altium), IR De-soldering (Chip-Off),
DevOps, Cloud & Virtualization: AWS; SaltStack, SCCM, MDT, Wazuh., Active Directory; Hyper-V, WSL, Ansible
Security, Networking & Automotive: pfSense, WireGuard; GNU Radio, can-utils, Ghidra, Wireshark, Nmap, Burp Suite, Ofrak, Trace32, GDB, OpenOCD, Volatility, Hashcat
Methodologies: Agile, CI/CD, Model-Based Systems Engineering

Professional Experience

Toyota Tsusho Systems, US — Plano, TX
Red Team Product Security Engineer | Embedded Security Engineer — 05/2026 - Present

- *Jidoka:* developed custom software for cybersecurity requirement verification on embedded automotive systems based on the ansible runner python API.
- *Vulnerability reporting:* internal security analysis and vulnerability research in automotive systems including: EV systems, Heavy-Duty systems, remote services, and cockpit ECUs.
- *AI tooling development:* MCP servers, skills, and internal local AI set up for reverse engineering and automation.

Colorado State University — Fort Collins, CO
Graduate Research Assistant | Embedded Security Engineer — 05/2024 - 05/2026

- *Ultimate Truck Hacking Platform:* Custom Yocto OS research with the NMFTA, used heavy vehicle network analysis.
- *CyberEvent SysAdmin:* Lead Network & IT Consultant for CyberX Challenges.
- *Patching ECMs using OFRAK:* Collaborated with Red Balloon Security on encrypting CAN bus communications.
- *OneNet Gateway:* Worked with the NMEA on developing the first NMEA 2K to NMEA OneNet protocol converter.
- *Electronic Logging Devices:* Reverse engineering mandated IoT logging devices and standards development work with the FMCSA.

Colorado State University — Fort Collins, CO
Engineering Technical Services | Support Technician — 08/2022 - 01/2024

Continuous Improvement

- *Licenses & certifications:* Bendix Brakes, GitHub Foundations, CSU Systems Security
- *Volunteering:* Black Hat MEA Car Hacking Village, CyberX Events, DEFCON34, Hackfesta
- *Hackathons & CTFs:* CyberSEED 2024 & 2025, SwampCTF 2024, MITRE eCTF 2025, MHV CTF 2025, USDA RibClassification, 2026 ToyotaSWARM Hackathon